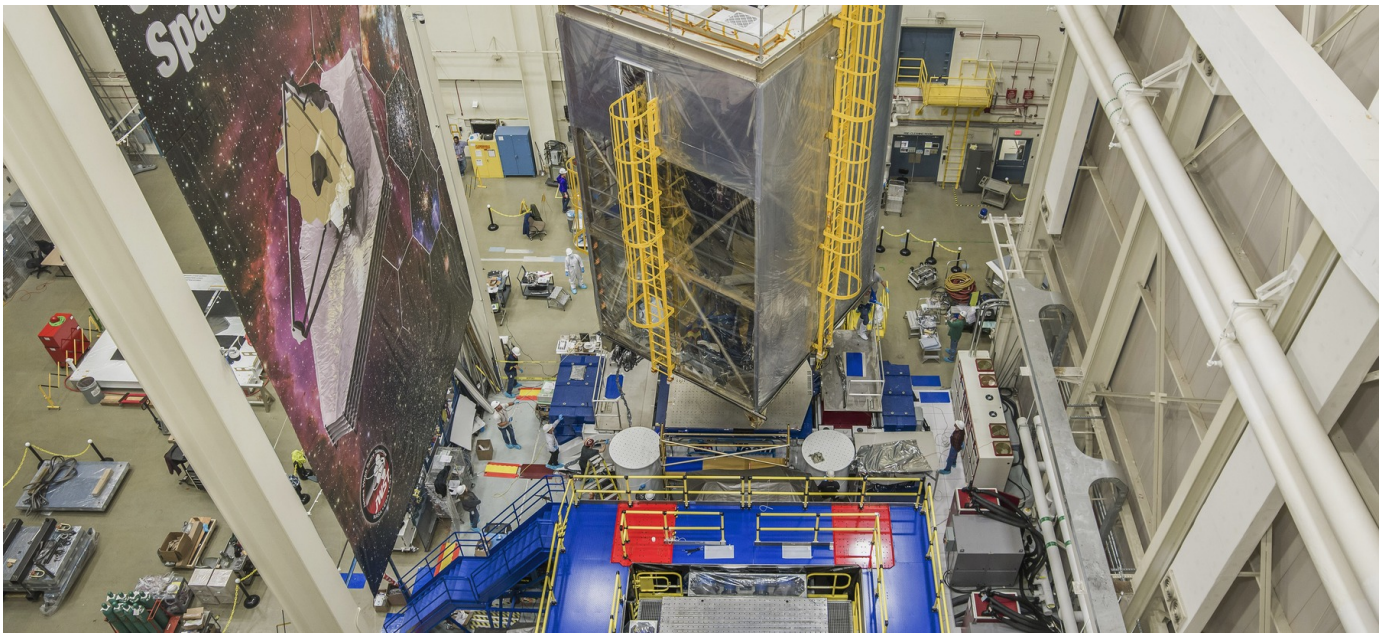
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Why a PSD trace needs error bars



Engineers and technicians perform vibration testing on the James Webb Space Telescope at NASA Goddard. The hardware is real; today's statistical examples are not Webb data. Photo: NASA/Chris Gunn, February 13, 2017; NASA source [4].

The line moved. Did the structure?

Imagine the morning after a vibration test. One response PSD has a new tooth near 220 Hz—about 2 dB higher than yesterday. It is tempting to tell a mechanical story: a mode shifted, a joint loosened, a controller found a resonance. But first ask a quieter question. How much can this PSD estimator move when the underlying random process has not changed at all?

That is today's objective: connect spectral averaging to effective degrees of freedom, turn those degrees of freedom into a confidence interval, and use the result without confusing estimator uncertainty with test tolerance or hardware margin. The payoff is practical. A trace with uncertainty attached is evidence; a trace without it is often only a picture.

A power spectral density is an ensemble quantity. The ideal definition asks for an expected value over infinitely many realizations. A test or flight gives us one finite record, so the plotted spectrum is an estimate. NASA's dynamics handbook makes this finite-record distinction explicit and notes that discrete estimates also carry frequency-resolution bias [3]. The jaggedness is not a software defect. It is the visible cost of estimating a statistical object from limited data.

One frequency bin, one random draw

For a stationary Gaussian random process, a raw periodogram ordinate is remarkably noisy. At an ordinary positive-frequency bin, its value behaves approximately like the true PSD multiplied by a chi-square random variable with two degrees of freedom, divided by two. The estimate is unbiased under the idealized assumptions, yet its standard deviation is as large as its mean. A longer unaveraged record gives finer bins; it does not automatically make each bin settle down.

Averaging changes the situation. Welch's 1967 method divides the record into shorter segments, applies a window, forms a modified periodogram for each segment and averages the results [1]. If the contributing estimates were independent and the resulting estimator had ν effective degrees of freedom, then at a selected frequency:

$$\frac{\nu \widehat{G}(f)}{G(f)} \sim \chi^2_\nu$$

Here G(f) is the unknown true one-sided PSD and Ĝ(f) is its estimate. The normalized scatter has coefficient of variation

$$\frac{\sigma_{\widehat{G}}}{G} \approx \sqrt{\frac{2}{\nu}}$$

With K independent periodograms, the familiar approximation is ν ≈ 2K. Four averages therefore imply about 50% relative standard deviation; sixteen imply 25%; sixty-four imply 12.5%. Those percentages describe bin-to-bin estimator scatter around a steady broadband level, not the uncertainty of a deterministic tone's amplitude.

The word effective matters. Overlap recovers data de-emphasized by a window, but adjacent segments become correlated. Window shape, overlap, spectral smoothing and even implementation details change the equivalent number of independent pieces of information. Counting segments and writing ν = 2K can overstate confidence unless independence is justified or effective DOF is computed for the actual estimator.

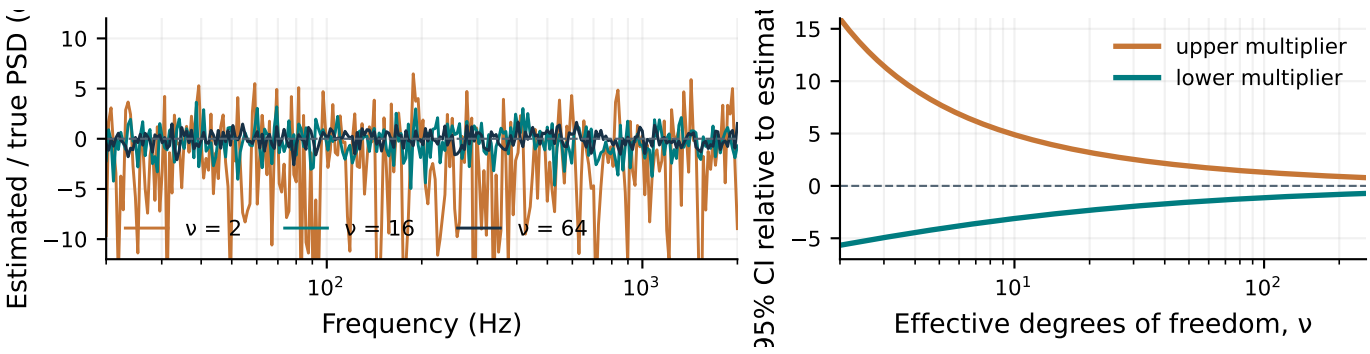


Figure 1. Original statistical illustration. Left: three seeded chi-square realizations about a flat true PSD; the apparent peaks are not simulated modes. Right: exact 95% chi-square interval multipliers. These are illustrative calculations, not measured test data.

Put bounds around the trace

Rearranging the chi-square relationship gives a two-sided 100(1−α)% confidence interval for the unknown PSD at one frequency. The quantile $\chi^2_{p,\nu}$ is exceeded with probability 1−p:

$$\frac{\nu \widehat{G}}{\chi^2_{1-\alpha/2,\,\nu}} \leq G \leq \frac{\nu \widehat{G}}{\chi^2_{\alpha/2,\,\nu}}$$

The asymmetry is real. A low-DOF estimate can sit well below the true PSD, so the upper multiplier stretches much farther than the lower multiplier. NIST gives the same chi-square interval structure for an unknown variance [2]; a PSD ordinate under the stated model is a frequency-resolved variance estimate.

Independent averages, K	Approx. ν	95% multiplier on Ĝ	Equivalent dB bounds
4	8	0.456 to 3.670	−3.41 / +5.65 dB
16	32	0.647 to 1.750	−1.89 / +2.43 dB
64	128	0.794 to 1.298	−1.00 / +1.13 dB

Table 1. Exact 95% chi-square multipliers for the independent-average illustration. PSD ratios use 10 $\log_{10}$, not 20 $\log_{10}$. Real windowed and overlapped estimates require effective DOF.

Now return to the 220 Hz tooth. Suppose an illustrative estimate is $\hat{G} = 0.10$ g²/Hz and the processing truly provides $\nu = 32$. The 95% interval for the underlying PSD is

$$G \in [0.647,\ 1.750](0.10) = [0.0647,\ 0.1750]\ \mathrm{g^2/Hz}$$

That is −1.89 to +2.43 dB relative to the plotted estimate. A 2 dB difference between two individual traces is therefore not automatically a structural change. It may still matter—especially if the feature is coherent across sensors, repeats, tracks a mode, or persists through more averaging—but the single-bin height alone is weak evidence at this DOF.

There is one more trap: this interval applies to one preselected frequency. If you scan hundreds of bins and report only the largest excursion, the chance of seeing at least one dramatic-looking point is much higher than 5%. Simultaneous bands, peak-persistence rules or a hypothesis chosen before looking are needed when the decision is “did anything change anywhere?”

The confidence band is not a permission slip

The chi-square result is powerful because it is simple, and dangerous for the same reason. It assumes a stationary, noise-like process and a PSD estimator whose effective degrees of freedom are known. Launch, separation and many machinery events are nonstationary. A PSD can still be a useful stationary-equivalent description, but a neat interval cannot restore time information that the averaging removed.

A narrow deterministic tone also asks a different question. Its apparent height depends on record synchronization, window coherent gain, leakage, bin spacing and whether the reporting quantity is line power, PSD or amplitude spectral density. Harris's classic window study catalogues the resolution and leakage trade space [5]. Applying broadband chi-square language to a migrating tone can make a processing artifact look like random uncertainty.

Then there are uncertainties the interval does not contain: accelerometer calibration and transverse sensitivity; clipping and electrical noise; anti-alias filtering; mounting and cable motion; controller dynamics; spatial variability; environmental run-to-run variability; and bias from a resonance narrower than the analysis bandwidth. Confidence in the estimator is not confidence in the entire measurement chain.

Nor is a 95% band a test tolerance. Controller tolerance bands, abort logic, notching rules and qualification levels come from the governing test specification and risk posture. NASA-HDBK-7005 is a technical handbook intended to support consistent practice; it is not a universal requirement that supersedes a program's approved procedure [3]. Statistical uncertainty should inform how a deviation is interpreted, not quietly rewrite acceptance criteria.

A useful diagnostic is to reprocess the same record several ways. Increase the number of averages and watch whether the feature stabilizes. Change segment length and check whether the energy remains localized or spreads with the window. Inspect time histories and spectrograms for stationarity. Look for phase or coherence evidence across channels. If the physical story is real, it should survive reasonable processing choices and appear in observables that the proposed mechanism actually connects.

When comparing two tests, uncertainty exists in both spectra. Two overlapping pointwise intervals do not constitute a formal equality test, and two non-overlapping intervals are not a complete root-cause argument. The engineering decision should combine estimator statistics with repeatability, sensor evidence, transfer-function behavior and the known configuration change.

Make the spectrum auditable

A PSD plot should travel with the metadata that set its statistical character: sample rate, record duration, segment length, window, overlap, detrending, averaging method, frequency resolution, normalization and effective DOF or confidence band. “Hann, 50% overlap” is not yet a complete uncertainty statement.

Begin a review with the decision being made. For a broad fatigue plateau, integrate over the physical band and assess uncertainty at that bandwidth. For a possible mode, preserve enough resolution to separate it from leakage and seek cross-channel evidence. Separate a controller’s display averaging from the evidence used for post-test interpretation.

Shorter segments and more averaging reduce variance but broaden features. Longer segments improve nominal resolution but leave fewer averages in a fixed record. Overlap can use data efficiently, although correlated segments do not buy a full pair of degrees of freedom each. Match the estimator’s resolution and uncertainty to the physical decision.

Three takeaways

1. A finite-record PSD is a random estimate; the trace should be read with its effective degrees of freedom.
2. Averaging narrows statistical scatter, but windows, overlap and smoothing determine the actual variance–resolution trade.
3. A confidence interval describes estimator uncertainty under a model. It is not a hardware margin, calibration budget or compliance tolerance.

Reflection: A response peak rises 2 dB after a configuration change. What should you ask before calling it structural?
Answer: Was the estimator and its effective DOF unchanged; is the rise larger than expected scatter; and does it persist across records, channels and processing choices consistent with the proposed mechanism?

Sources, scope and figure credits

[1] P. D. Welch, “The Use of Fast Fourier Transform for the Estimation of Power Spectra,” IEEE Transactions on Audio and Electroacoustics, 15(2), 70–73, June 1967. DOI 10.1109/TAU.1967.1161901.

[2] NIST/SEMATECH e-Handbook of Statistical Methods, “Chi-Square Test for the Variance,” including the two-sided interval form. Accessed October 8, 2026.

[3] NASA-HDBK-7005, Dynamic Environmental Criteria, March 13, 2001, especially §2.2.4 on finite-record autospectral estimates. Handbook guidance; applicability is controlled by the relevant program or contract.

[4] NASA Webb Telescope Team, “NASA Gives the Webb Telescope a Shakedown,” February 13, 2017; photo NASA/Chris Gunn.

[5] F. J. Harris, “On the Use of Windows for Harmonic Analysis with the Discrete Fourier Transform,” Proceedings of the IEEE, 66(1), 51–83, January 1978. DOI 10.1109/PROC.1978.10837.

Figure 1 and Table 1 were created for this lesson. Calculations are illustrative and reproducible from the cited chi-square model; they are not Webb measurements or qualification criteria. Sources accessed October 8, 2026. GPT-6 Astra selection could not be verified for this edition.